



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2014
Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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H2 BIOLOGY

9648/02

Paper 2 Core Paper

18 September 2014

2 hours

Additional Materials: Answer paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough workings.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination,

1. Fasten your answer papers to section B securely together.

2. Hand in the following separately:

- Section A (Part I)
- Section A (Part II)
- Section B

The number of marks is given in brackets [] at the end of each question or part question.

For examiner's Use	
Section A	
1	/ 10
2	/ 10
3	/ 9
4	/ 11
5	/ 10
6	/ 10
7	/ 10
8	/ 10
Section B	/ 20
Total	/ 100

Section A
Answer **all** questions in the spaces provided.

For
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- 1** The small intestine is lined with thousands of small finger-like projections called villi. Each villus is covered by a layer of cells and this lining of cells is called the epithelium.

In an investigation to study the cell cycle of epithelial cells, radioactive-labelled thymine was used. The radioactively labelled thymine was injected into the blood of a mammal. Some of this radioactive thymine was absorbed from the blood by the epithelial cells.

- (a)** The radioactively labelled thymine only appeared in the cells 0.5 hours after injection.

- (i)** Suggest why this is so. [1]

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- (ii)** Describe how radioactively labelled thymine enter the cells. [2]

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- (b)** Suggest why radioactive thymine was used to label the DNA in these cells rather than radioactive cytosine. [2]

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Using the radioactive-labelled DNA, a cell with condensed chromosomes was carefully observed and the following measurements were taken:

A – distance from pole to pole

B – distance between a centromere and the respective poles

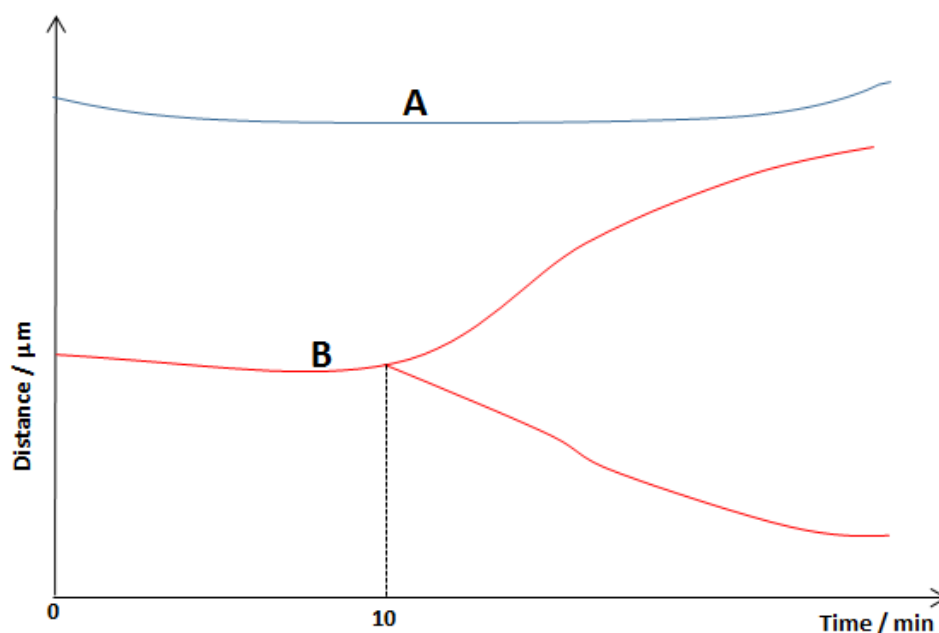


Fig. 1.1

(c) Describe the event that occurred in B after 10 minutes. [3]

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(d) Some drugs which are used in the treatment of cancer affect the function of spindle fibres. Suggest what would happen if the drug is introduced into the cell at 0 min. [2]

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[Total: 10]

2 Fig. 2.1 shows a molecule of transfer RNA (tRNA) attached to a corresponding enzyme.

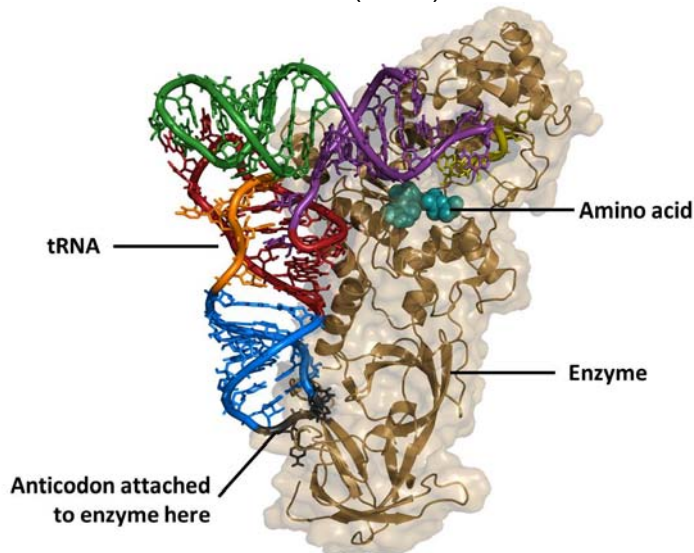


Fig. 2.1

(a) Identify the enzyme labelled in Fig. 2.1. [1]

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(b) Describe how the structure of the tRNA molecule and the enzyme molecule is maintained. [2]

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(c) A specific example of the enzyme in Fig. 2.1 allows the amino acid histidine to bind to its active site. A single base substitution occurred in the gene coding for this enzyme results in the enzyme recognizing tyrosine instead of histidine.

Suggest why a mutation in this gene would have extremely damaging effects on the organism. [3]

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(d) The flow of genetic information from DNA to polypeptide is an accurate process. Explain how this is so. [4]

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[Total: 10]

3 The *c-myc* proto-oncogene on chromosome 8 codes for the c-myc protein, a transcription factor that promotes cell proliferation. In cells that are induced to differentiate, the gene is expressed at a very low level. Fig. 3.1 shows the involvement of two enzymes in regulating the expression of this gene.

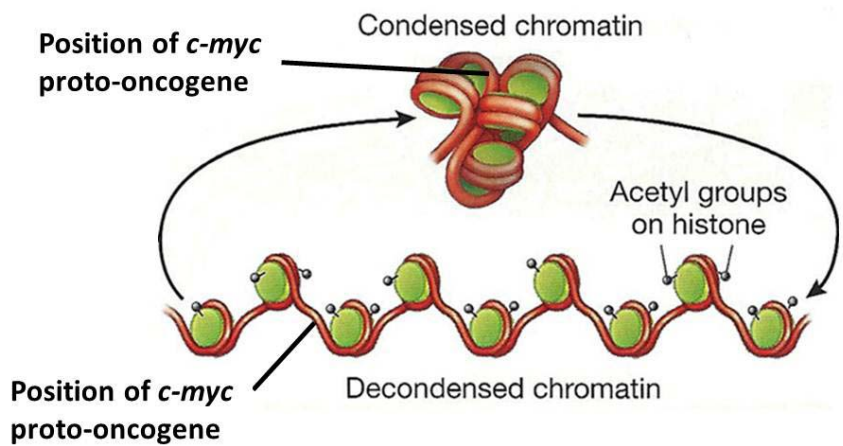


Fig. 3.1

- (a)** Using information from Fig. 3.1 and your knowledge on chromatin remodelling, explain how the expression of the *c-myc* proto-oncogene is repressed. [4]

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- (b)** The *c-myc* protein, as a transcription factor essential for cell proliferation, activates a wide range of cellular activities. One of which is to increase the rate of transcription mediated by all three RNA polymerases, I, II, and III. These different types of RNA polymerases are essential for the synthesis of rRNAs, 5S RNA, and ribosomal proteins.

- (i)** State the stage of the cell cycle that this occurs. [1]

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- (ii)** Explain why this is important to cell proliferation. [2]

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- (c)** An oncovirus is a virus that can cause cancer. In one class of oncovirus, the viral genome is inserted near the *c-myc* proto-oncogene in the host genome.

Suggest how this process would lead to the development of cancer. [2]

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[Total: 9]

- 4 The activity of the *lac* operon in three different *E. coli* strains was measured and the results shown in Table 4.1. The *lac Z* gene codes for the enzyme β -galactosidase. The *lac Y* gene codes for β -galactoside permease.

<i>E. coli</i> strain	Activity of <i>lac I</i> protein	Expression of protein coded by <i>lac Z</i>	Expression of protein coded by <i>lac Y</i>
1 (normal)	+	+++	++
2 (mutant)	+	-	-
3 (mutant)	+	+++	-

Table 4.1

- (a) Describe the reaction catalysed by β -galactosidase. [2]

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- (b) In the presence of only lactose and assuming that only one mutation occurred in the operon,

- (i) explain the results for *E. coli* strain 2; [3]

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- (ii) explain the results for *E. coli* strain 3 and its consequences to the cells. [3]

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(c) Explain why having both inducible and repressible operons is advantageous to the prokaryotic cell. [3]

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[Total: 11]

- 5 The capsules of shepherd's purse occur in two different shapes, i.e., triangular and top-shaped. When a pure breed plant with triangular capsule is crossed with another pure breed plant having top-shaped capsule, all F_1 have only triangular character.

The F_1 offspring were selfed and the following F_2 results was obtained:

1468	triangular-shaped capsules
87	top-shaped capsules

- (a) Using suitable symbols, draw a genetic diagram to explain the results of the F_2 generation. [4]

(b) Explain the genetic basis of this type of inheritance. [3]

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The chi-squared (χ^2) test was performed on the F_2 data giving a calculated value for χ^2 of 2.73.

Degrees of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Table 5.1: Distribution of χ^2

(c) Using the information given thus far,

(i) determine the degree of freedom [1]

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(ii) state what conclusion may be drawn regarding the observed F_2 data. [2]

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[Total: 10]

6 The following experiment was carried out to investigate the effect of light intensity on the rate of photosynthesis of a water plant, *Elodea*.

- *Elodea* was cut into three pieces, each 10 cm long.
- Each piece of *Elodea* was placed in a glass tube, containing 0.5% sodium hydrogen carbonate solution, which was then sealed with a bung.
- Tube **A** was placed 10 cm away from a lamp.
- Tube **B** was placed 5 cm away from a lamp.
- Tube **C** was placed in a dark room.
- An oxygen sensor was used to measure the percentage of oxygen in the solutions at the start of the experiment and again at 5, 10 and 20 minutes.

The results are shown in Fig. 6.1.

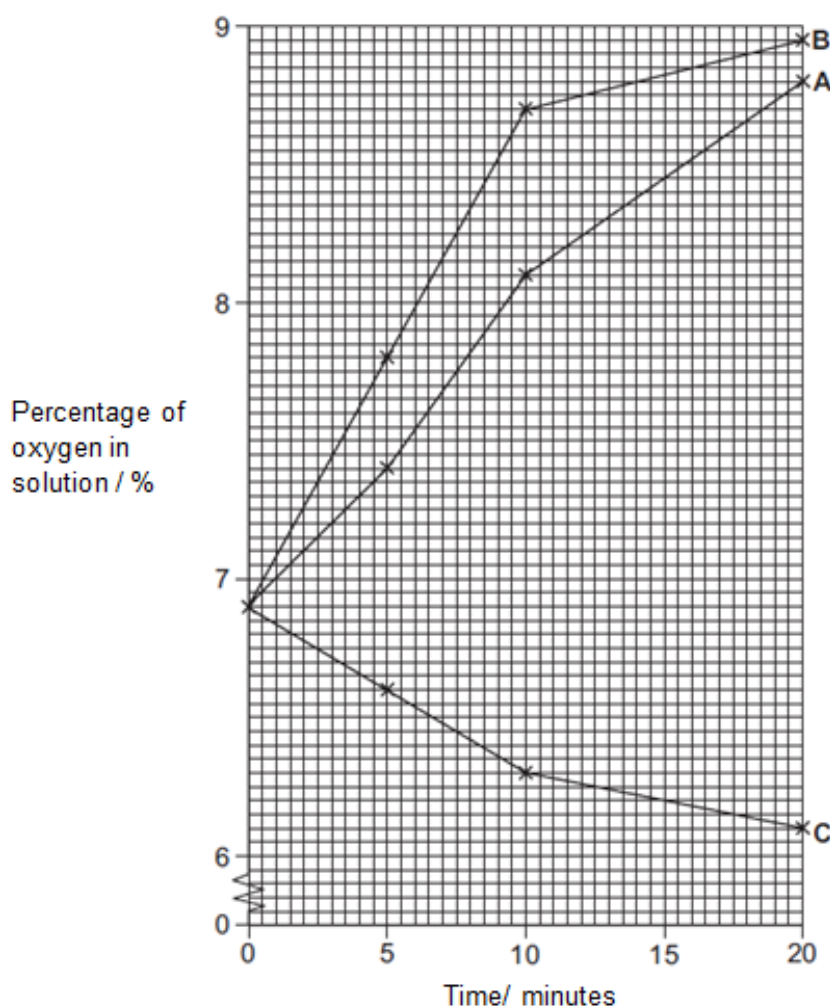


Fig. 6.1

(a) With reference to Fig. 6.1,

- (i) Calculate the rate of oxygen production for tube **A** for the 20 minutes of the experiment. Give your answer to two decimal places. Show your working. [1]

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(ii) Compare the results for tubes **A** and **B**. [2]

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(iii) Explain the results for tube **C**. [2]

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(b) Fig. 6.2 outlines the early stages of respiration.

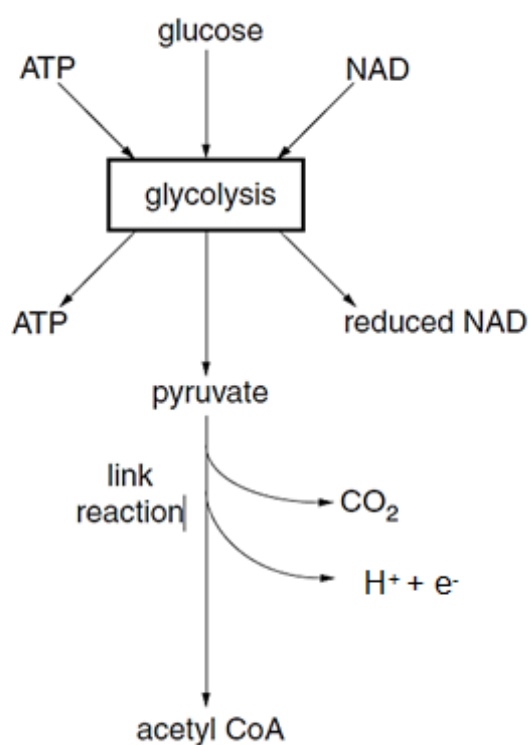


Fig. 6.2

With reference to Fig. 6.2,

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(i) explain why ATP is needed at the start of glycolysis. [1]

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(ii) explain the fate of the hydrogen atoms released by pyruvate during the link reaction. [4]

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[Total: 10]

- 7 In MAPK (Mitogen Activated Protein Kinase) pathway, epidermal growth factor (EGF) hormones circulating in the blood are able to trigger transcription within a cell, even though they are unable to enter the cell. This eventually results in genes switching on and transcription beginning.

Fig. 7.1 shows part of the MAPK pathway.

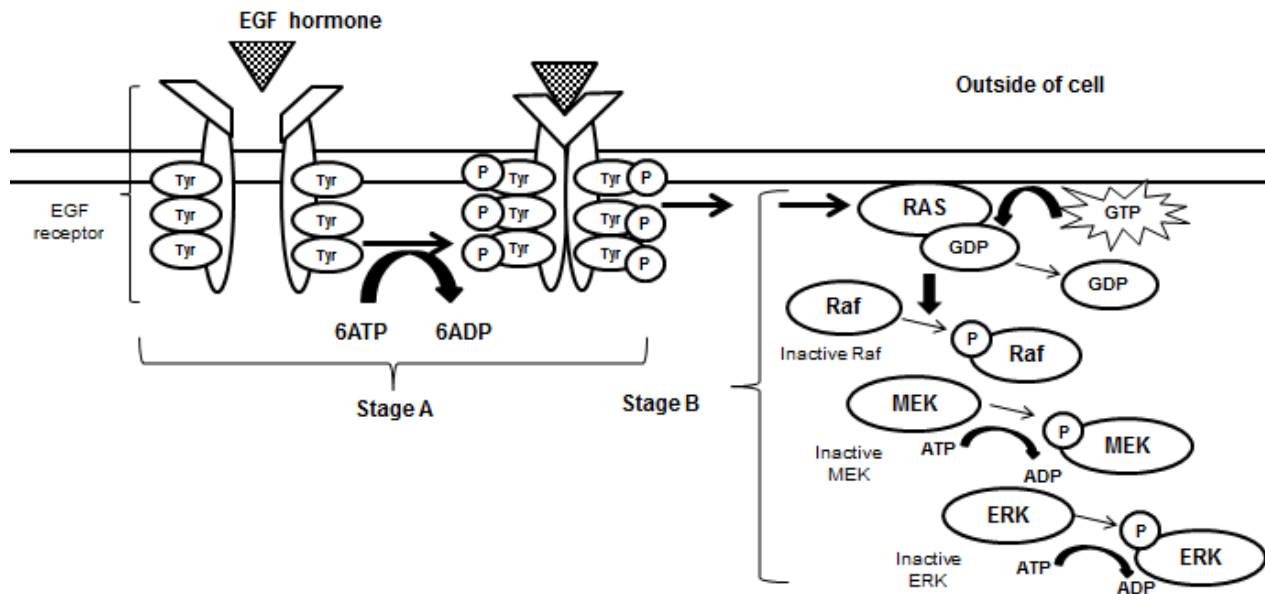


Fig. 7.1

- (a) (i) Explain why the EGF hormone is unable to enter the cell. [2]

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(ii) Describe the stages A and B. [4]

Stage A:

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Stage B:

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(iii) Suggest how the signal transduction pathway of MAPK may be turned off in Stage B. [1]

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(b) Describe the differences in an organism’s responses to neural signalling and endocrine signalling. [3]

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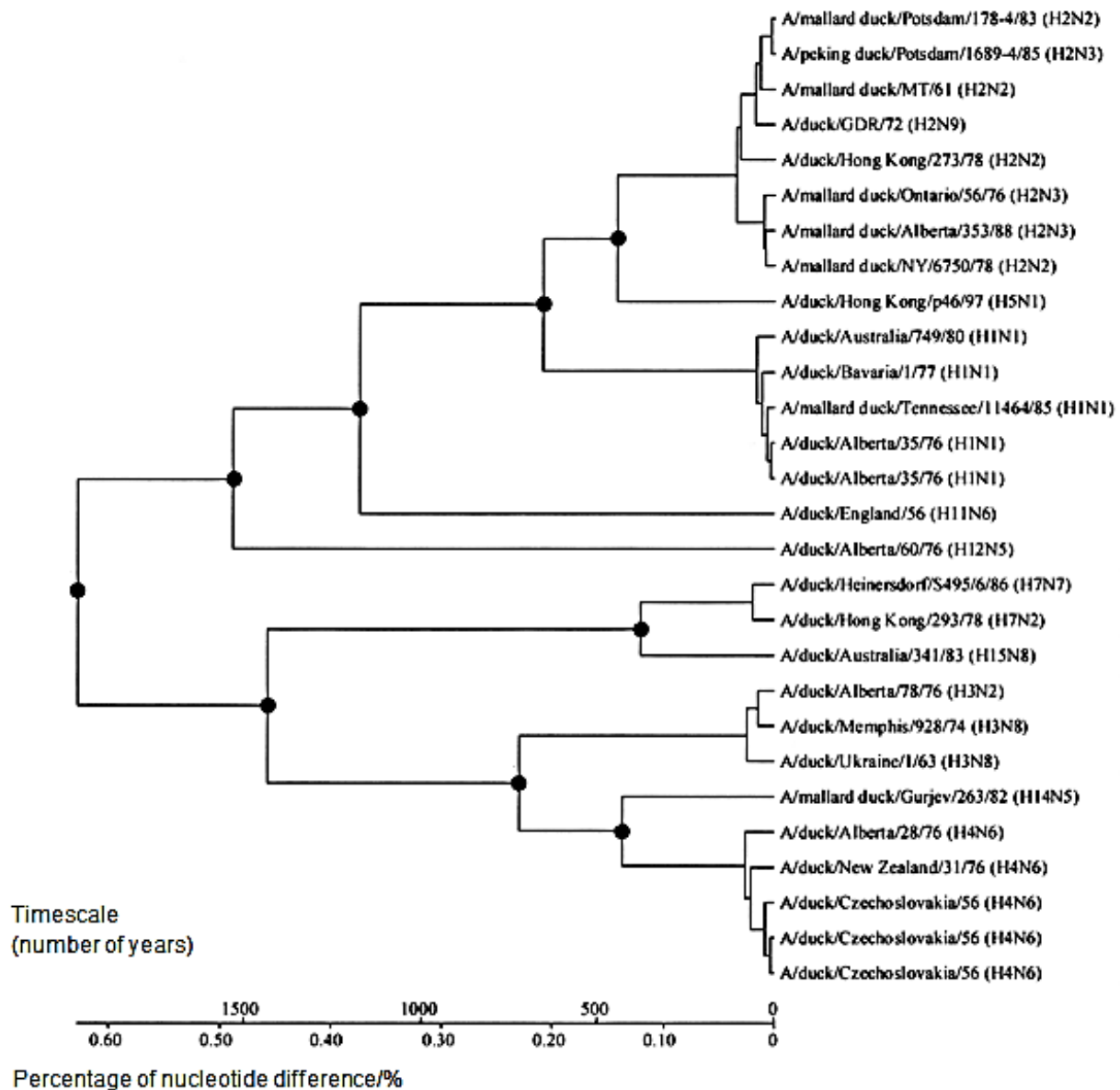
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[Total: 10]

- 8 Fig. 8.1 shows the phylogenetic tree of different duck influenza A virus constructed from studying the nucleotide sequence of haemagglutinins (HAs). The black circles denote nodes at which the divergence times were estimated. An ancestral Influenza virus evolved into strains that infected ducks. Over time, new strains began to infect humans and swine.



Source: <http://mbe.oxfordjournals.org/content/19/4/501.full#T4>

Fig. 8.1

- (a) Explain the relationship between classification and phylogeny. [2]

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(b) With reference to Fig. 8.1, describe how the phylogenetic tree is constructed. [3]

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Influenza A virus causes epidemics of influenza in human. It appears to occur when new subtypes of influenza virus A are introduced from aquatic birds into human.

(c) Explain how influenza virus A from duck is able to infect human. [3]

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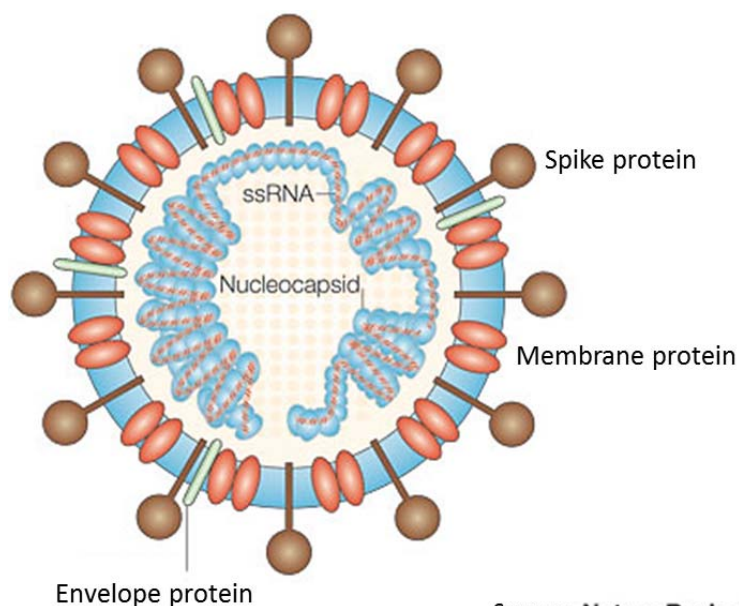
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Coronaviruses are enveloped RNA viruses that infect and cause lower respiratory tract disease in a broad array of animals and human. Virus particles range from 70 to 120 nm in diameter and are surrounded by characteristic spike-shaped glycoproteins, as shown in Fig. 8.2. Coronaviruses contain single-stranded RNA genomes that can be used directly for translation.



Source: **Nature Reviews | Immunology**

Fig. 8.2

(d) Describe **two** differences between the genome of the coronavirus and influenza virus. [2]

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[Total: 10]

Section B

Answer **one** question

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Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in questions **(a)**, **(b)**, etc., as indicated in the question.

QUESTION 9

- (a) State the significant chemical features of phospholipid and explain its biological importance. [7]
- (b) Outline the similarities and differences between homologous chromosomes. [5]
- (c) Discuss the effect of inhibitors on rate of enzymatic reaction and how it can be overcome. [8]

[Total: 20]

QUESTION 10

- (a) Describe how the involvement of bacteriophages increases genetic variation of a bacterial colony. [7]
- (b) Outline the similarities and differences between a secretory vesicle and a lysosome. [5]
- (c) Discuss the roles of telomeres and telomerase in enabling the function of stem cells. [8]

[Total: 20]

~ End of Paper ~